Automotive Vision Capture Plugin for Instrument Studio (PXIe-148X)

This plugin provides an easy-to-use **automotive camera (FPD-Link/ GMSL) acquisition, display, logging** workflow in **Instrument Studio 2025**, powered by NI FlexRIO **PXIe-148X** modules (e.g., **PXIe-1486**).

Contents

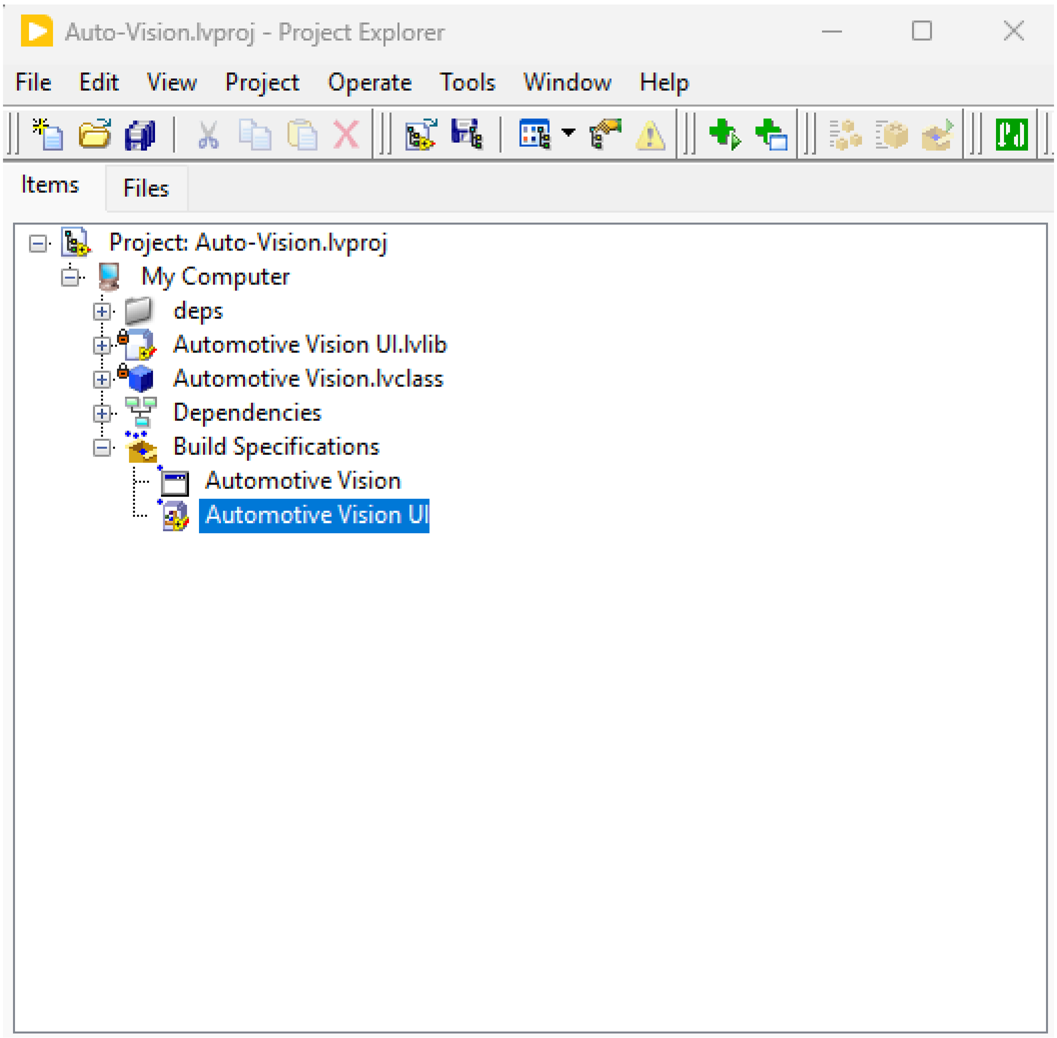
- **Service logic main VI**: `Automotive Vision\Measurement Logic.vi`
  Implements CSI-2 packet acquisition, image decode and logging using FlexRIO bitfiles and NI Vision VIs.
- **UI VI**: `Automotive Vision UI\Measurement UI.vi`
  Instrument Studio panel with controls for channel selection, payload/interpretation and image algorithms; includes status/telemetry (FPS, frames, PoC V/I).

Prerequisites

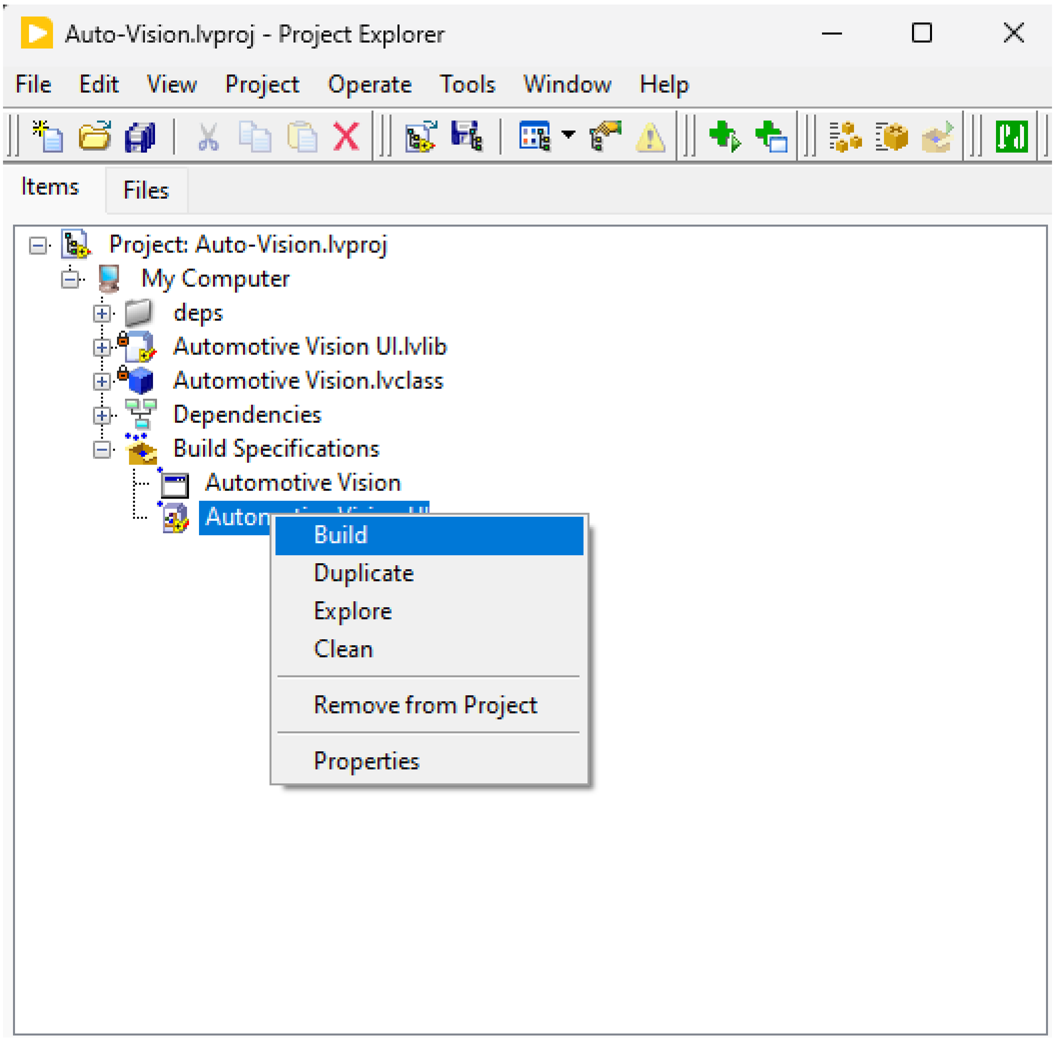
- **LabVIEW 2025** with:
  - **NI-FlexRIO** driver (23Q1 or newer) and **PXIe-148X** example set.
  - **Vision Development Module** (IMAQ/IMAQdx for image handling & decoding).
- **Instrument Studio 2025**.
- Supported **PXIe-148X** module (e.g., **PXIe-1486 8-In or 4-In/4-Out**) with camera and FAKRA coax cabling; PoC capable power as required.

How to Build & Run the Plugin

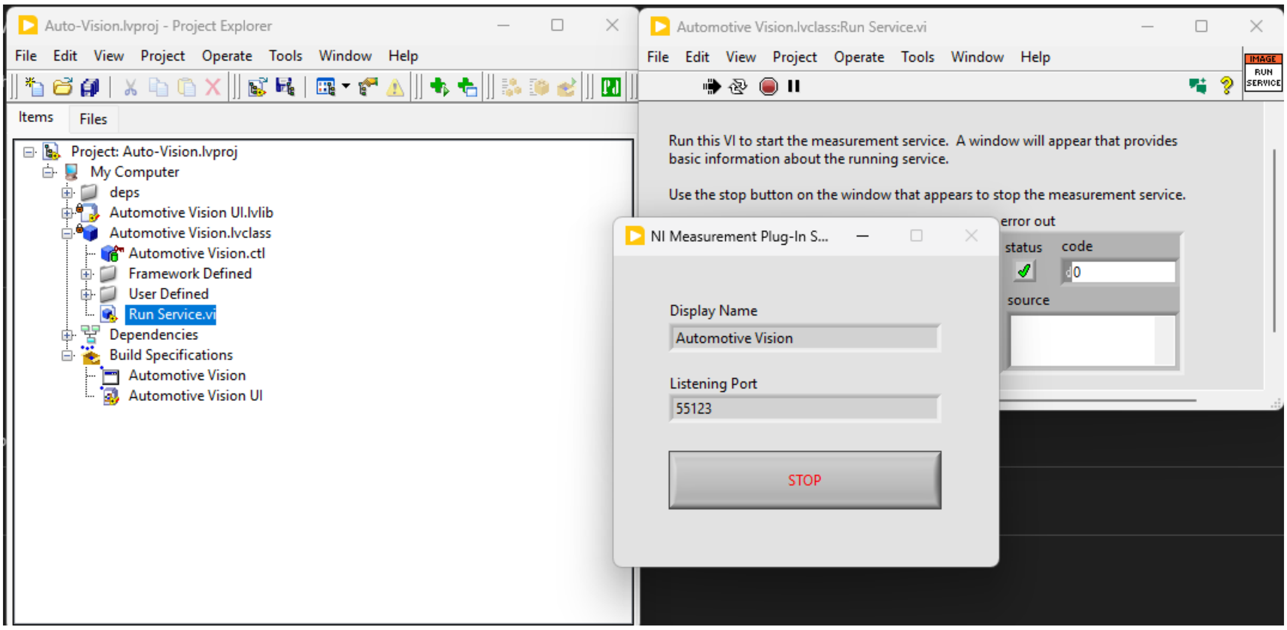
1. **Open project**
   Launch `Auto-Vision.lvproj` in **LabVIEW 2025**.



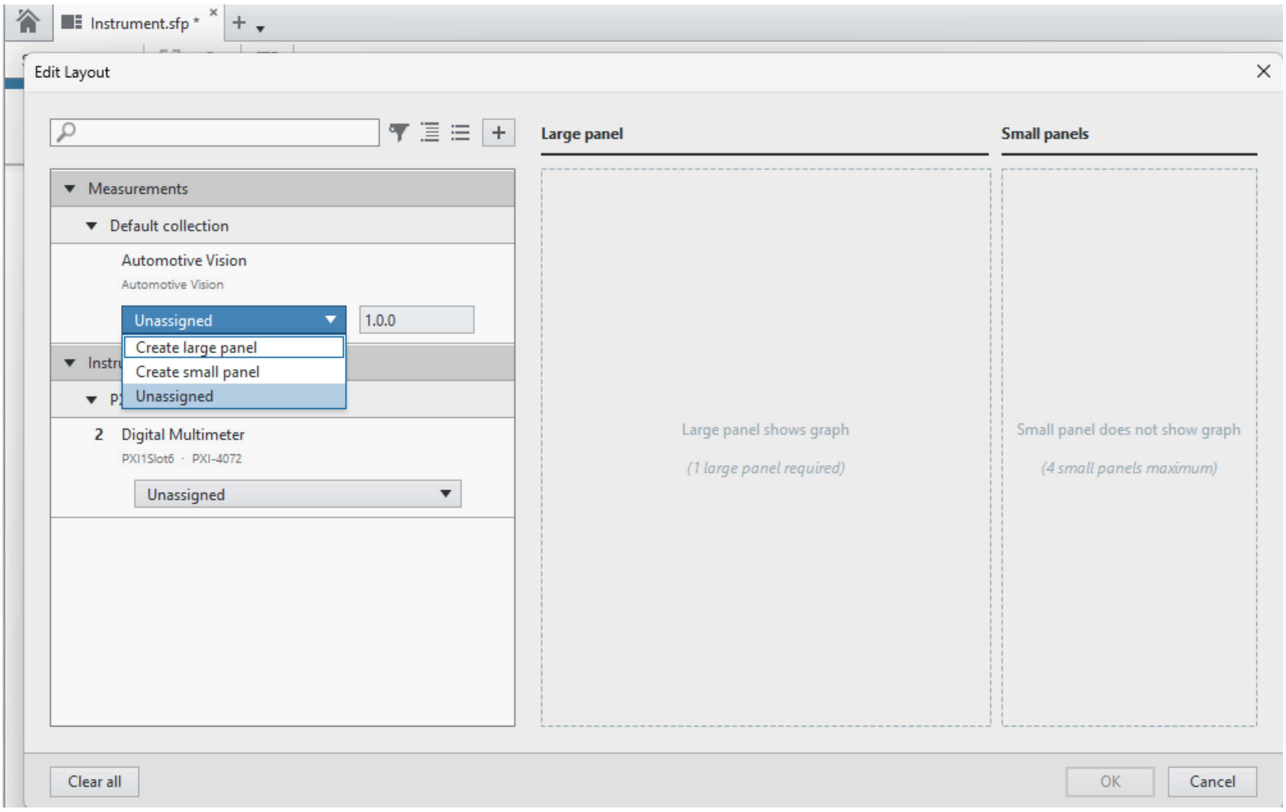
2. **Build the UI and the logic**
   Build the `Automotive Vision UI` and `Automotive Vision` specification (Application Builder).



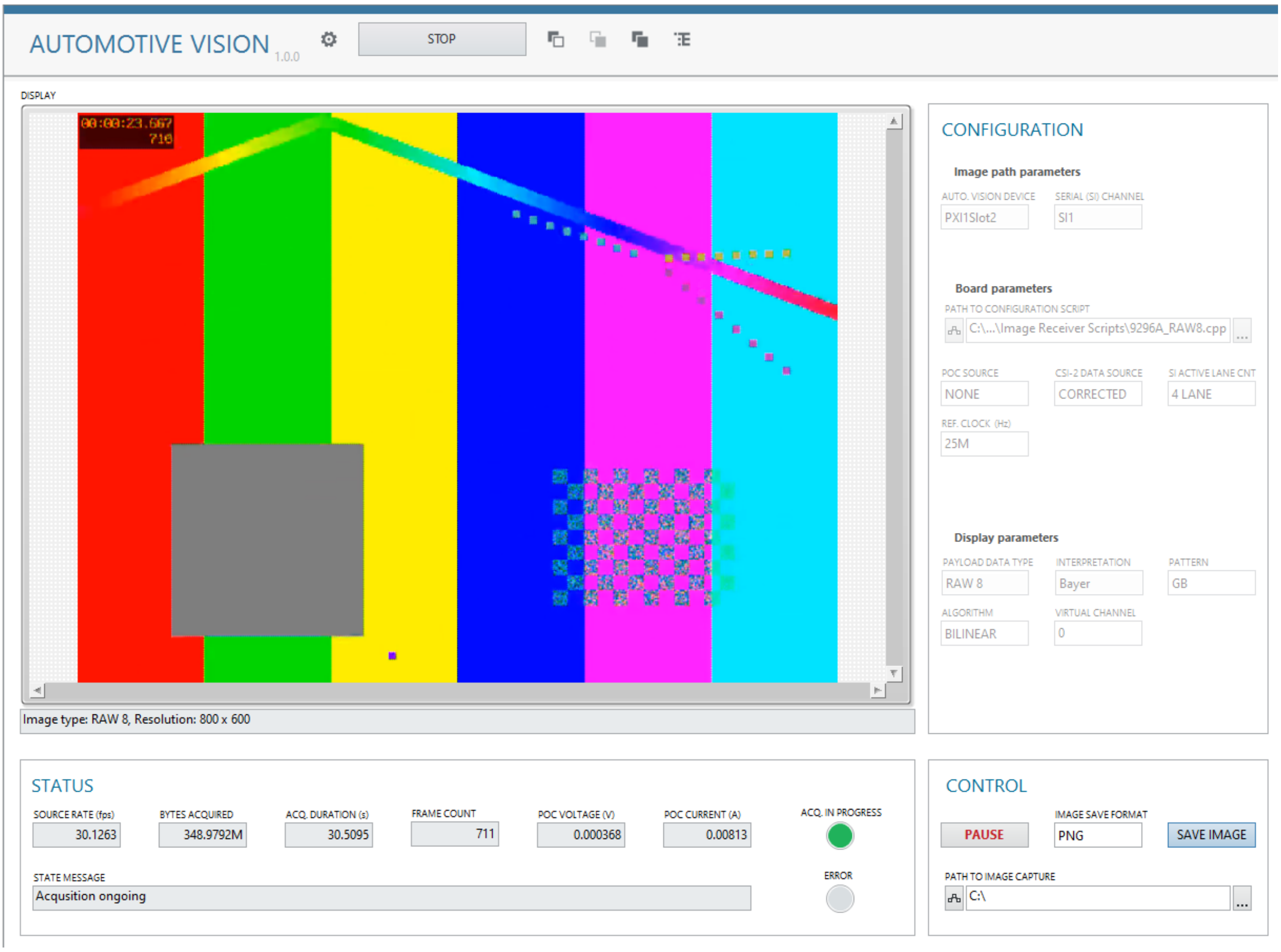
3. **Run the service**
   Open `Run Service.vi` in the **Automotive Vision** class and run.



4. **Open Instrument Studio (IS) 2025**
   Create a **Manual Layout**, add a **Large Panel** and select **Automotive Vision Capture**.

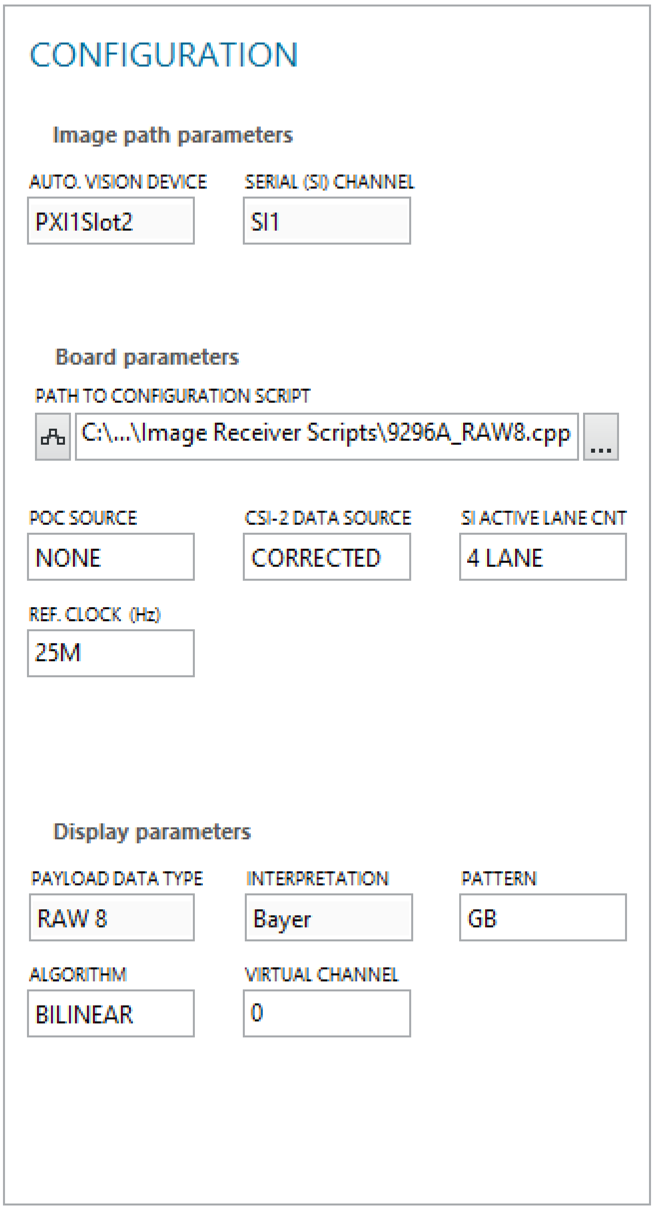


5. **UI in IS**
   The plugin panel appears with live image, packet/trace tabs and status telemetry.



Configuration

> These controls map to NI's PXIe-148X acquisition example front panel. Use values matching the **camera configuration script**.



CSI-2 Data Source

Select **RAW** or **Corrected** payload stream as provided by the camera/deserializer path.
TIP: RAW = sensor output; Corrected = post-processed/error-corrected.

Active Lane Count

Choose **1 / 2 / 4 lanes** for CSI-2 throughput. Higher lane count → higher bandwidth for high-res/FPS.
(PXIe-1486 supports 4-lane CSI-2, 800 Mbps/lane.)

Reference Clock Source

Set the **refclk** used for link sync (typically **25 MHz**, camera-dependent; input channels commonly **23–26 MHz**). Ensure it matches camera PLL settings.

Payload Data Type

Select pixel format (**RAW8**, **YUV422**, **RGB888**, etc.), exactly as configured in the **camera’s script**, so the decode/display matches the stream.

Interpretation

Choose how payload is decoded for display: **grayscale**, **Bayer**, **RGB**, **YUV420/422**. This must match the **data type** and the **camera output**.

Pattern (Bayer only)

If **Bayer** is selected, set the CFA pattern: **RG / BG / GR / GB** to reconstruct correct colors.

Algorithm (Bayer demosaicing)

Pick **Bilinear** (fast) or **VNG** (higher edge quality, slower). NI recommends Bilinear first; use VNG where edge fidelity matters.

Virtual Channel

Select the **CSI-2 Virtual Channel ID** to demultiplex multiple streams sharing the link. Useful for multi-sensor aggregates.

PoC (Power-over-Coax)

- Choose **PoC source**, observe **PoC voltage/current** indicators for camera power health.

Configuration — ** Board **

RIO Device & Bitfile

- Bitfile selected automatically based on your hardware selection.

AUTO VISION DEVICE

- Select which device to use.

SERIAL (SI) CHANNEL

- Select which serial input channel to use - only one channel is supported at a time.

Starting & Using the Plugin

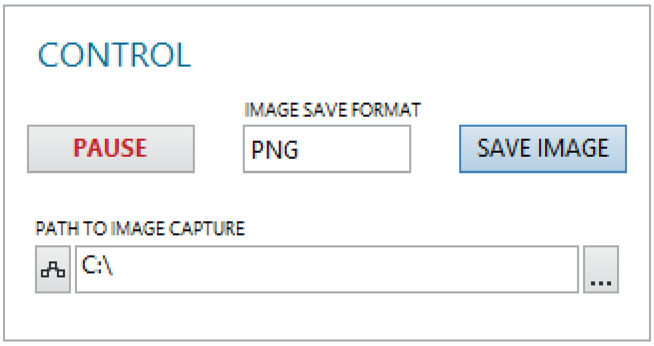
1. **RUN** the plugin in Instrument Studio.
   It connects to the selected serial channel(s), starts acquiring CSI-2 packets and image frames; status LED shows connection. The service updates the UI periodically (based on FPS). On higher resolution the FPS can drop.

2. **View live image & packets**

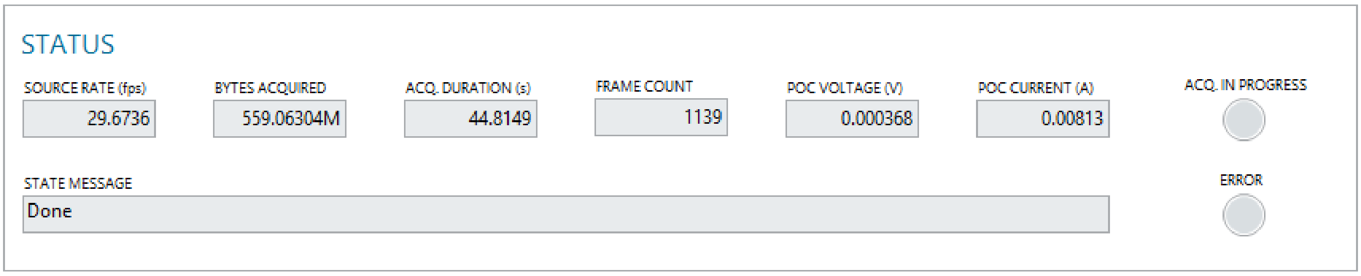
   - **Image pane** shows decoded frames.

3. **Interact via controls**

    - **Pause / Resume** (OOB gRPC) to suspend UI updates without stopping acquisition.
    - **Capture**: save one frame correspond to the settings above.
    - **Capture type**: can be PNG or RAW.
    - **Capture path**: a folder to save the capture - .bin or .png file.



Indicators (Status Telemetry)



- **Source rate (FPS)** – current frame rate per channel.
- **Bytes acquired** – total data captured this session.
- **Acquisition duration (s)** – elapsed time.
- **Frame count** – total frames received.
- **PoC voltage / current** – camera power telemetry over coax.
- **Acquisition in progress** – Indicates that the AutoVision device is capable of receiving frames.
- **Error** – Indicates error in measurement logic.

Tips & Known Behaviors

- **Match scripts to UI**: Payload type, interpretation, Bayer pattern and VC must align with the **camera configuration script**, else display/packet parsing will be wrong.
- **Throughput**: CSI-2 lane count and refclk affect bandwidth. PXIe-1486 supports **4-lane CSI-2 @ ~800 Mbps/lane** per NI docs.
- **Display channel count**: Default host/FPGA code displays up to **one** channels; more requires code edits.
- **Displaying performance**: Displaying **very large resolution or high FPS** may cause the VI to appear unresponsive while processing. Displayed FPS are reduced in these case.
- **PoC safety**: Monitor **PoC V/I**; ensure proper termination and power budgets for the sensor.
